CRYSS: a Data Analysis System for High-Throughput Crystallization Screening

CRYSS is an open-source insight generator for partitioned equilibrium datasets, designed to convert complex multicomponent solubility behaviour into mechanistic, decision-ready purification models.

Executive Summary

Introduction

Chemical synthesis almost always produces side products, making purification an essential part of process development. In multi-step pharmaceutical synthesis, even high step yields accumulate into substantial overall loss: a ten-step sequence at 90% per step delivers only ~35% overall yield (0.9^{10}). In practice, synthesis is an exercise in impurity removal.

Crystallization is one of the most attractive purification methods when the target molecule is a solid. It is operationally simple, scalable, and cost-effective. Yet optimizing a crystallization process is non-trivial: one must maximize space-time yield while meeting stringent quality targets. This involves controlling polymorphs, managing crystal growth to avoid defects and impurity inclusion, understanding terminal solid-solution behaviour, and applying strategies such as seeding or carefully designed cooling trajectories (or other routes to metastable supersaturation such as evaporation or antisolvent addition). These optimizations operate in a kinetically controlled regime and require significant effort.

Before any of that work begins, however, the primary screening question is simple:

Which systems give the highest selective removal of impurities, and therefore the highest achievable yield of the target molecule?

Answering this requires exploring solvents, solvent combinations (including proxies for antisolvent addition), and—where chemistry allows—derivatization. Salt formation is particularly attractive compared with covalent derivatization. These choices introduce categorical variables, and combinatorial complexity escalates quickly.

Parallel small-reactor blocks with stirring and thermal cycling provide an efficient way to screen purification behaviour across many systems. The challenge then becomes: **how do we interpret the results of such a screening program?**

CRYSS Workflow Overview

At its simplest, the method consists of the following steps (see Figure 1):

1) Equilibrate the candidate mixture (which may include different salt forms) with various solvents or solvent mixtures in a closed system to avoid solvent loss. Achieving equilibrium is essential, so thermal cycling, agitation, and sufficient time are required.
2) Determine the mass fraction of the mixture in solution and solid phases at equilibrium. If performed carefully, only calibrated analytics of the solution phase are needed: knowing the initial mass and the dissolved mass immediately gives the solid mass by difference. Avoiding solvent loss is critical, as evaporation artificially inflates the apparent dissolved

mass. Alternatively, separating solid and solution and applying area % analysis yields the same information via the lever rule, though this is more difficult at small scale.

3) Record each experiment in an Excel file, including:
    a. starting mixture composition (x_c),
    b. solid phase composition (x_s),
    c. solution phase composition (x_l),
    d. experiment or mixture ID.
4) Feed the dataset into CRYSS, which ranks each experiment by the predicted maximum achievable yield (R_{max}) once optimal process conditions are applied. CRYSS identifies the systems worth further development.

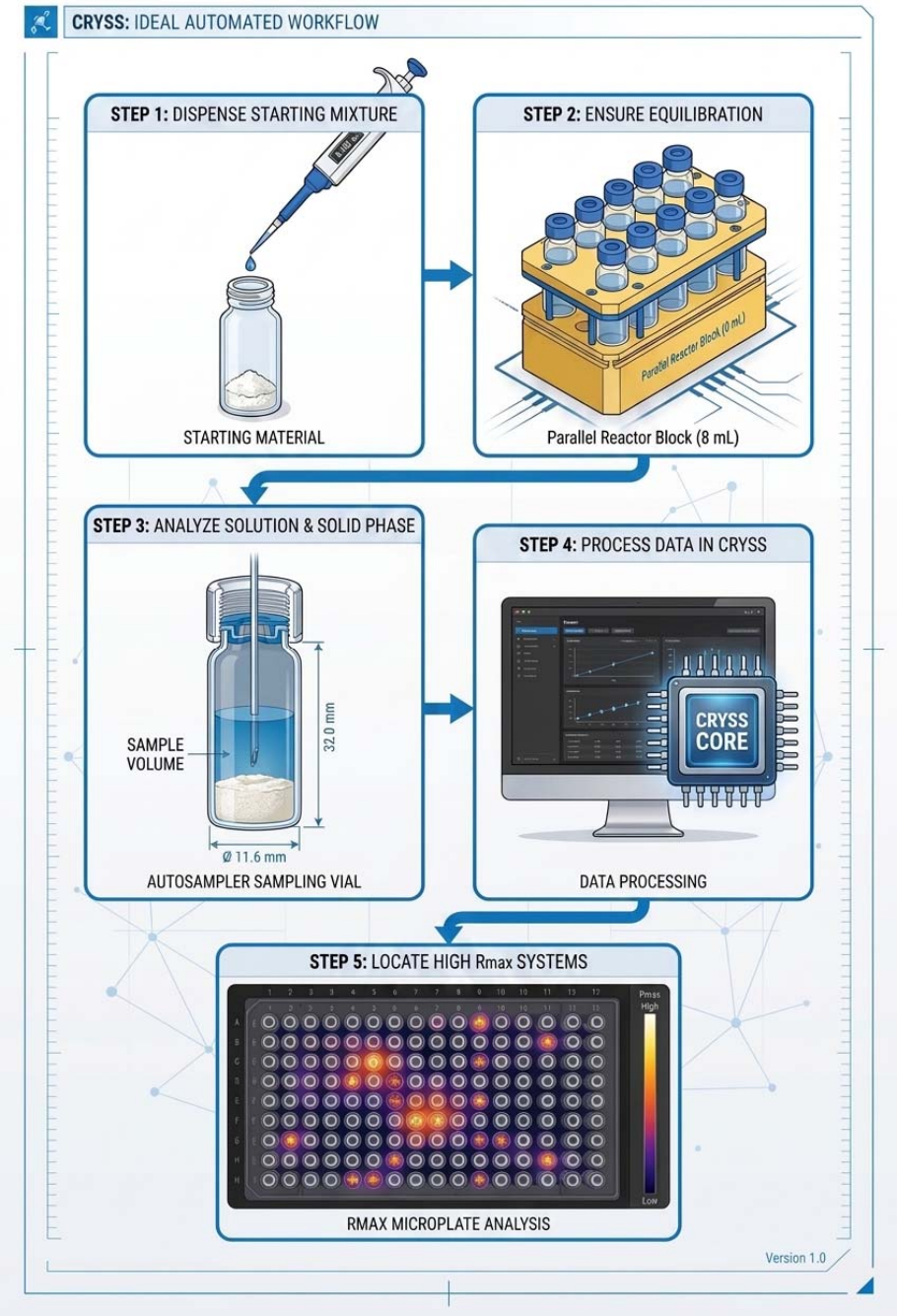


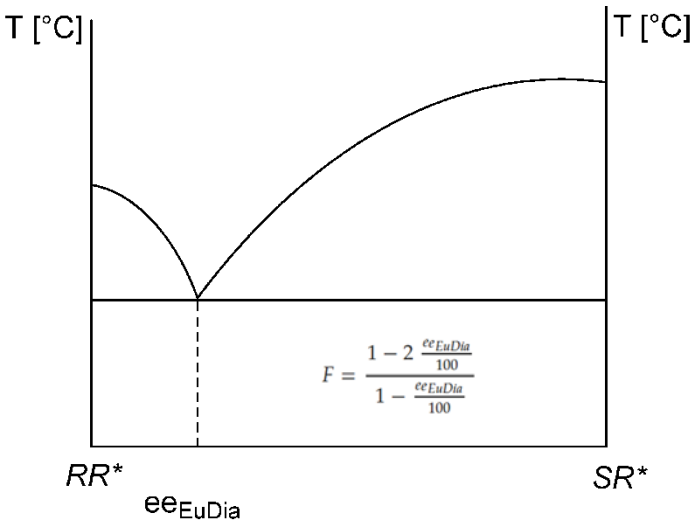
Figure 1: CRYSS enables a highly automated and consistent workflow

Core Principle

1) Binary System

In a binary system (e.g., diastereomeric salt resolution), if both components are present in the solid at equilibrium, the solution phase must have the eutectic composition (x_{eu}). For diastereomer mediated resolution of a racemate, knowing x_{eu} allows straightforward calculation of the maximum possible recovery of pure material:

$$R_{max} = \frac{(0.5 - x_{eu})}{(1 - x_{eu})}$$



This relationship is familiar from purification of enantioenriched systems.[1] However, multicomponent systems behave differently: the solution phase only has the true eutectic composition at R_{max}, not throughout the entire recovery range.

2) Multicomponent Systems

In multicomponent mixtures, solubility behaviour is encoded in how each component's level changes as a function of equilibrium partitioning at different recovery points R (see Figure 2). For example, if 1 g of mixture is equilibrated with a given solvent volume and 0.1 g dissolves, then 0.9 g remains in the solid phase: R = 90% recovery. R_{max} is the recovery corresponding to the maximum physical yield of pure material, and achieving it requires an optimal solvent volume. Using more solvent than this reduces yield.

Crucially, all information needed to identify a good system is encoded in the analytical profiles across the range R = 100% → R_{max}. This is analogous to diastereomer screening, but requires more sophisticated approach; CRYSS provides exactly that.

[1] Based on the melting binary phase diagram of diastereomeric mixtures, Ács et al. calculated that, based on the knowledge of the eutectic composition of the binary melting phase diagram, it is possible to calculate the resolvability (F) of the crystalline precipitating diastereomer of the given racemic compound with the given resolving agent. This is a generally applicable method if, after optimization, diastereomeric mixtures have been obtained, and their melting binary phase diagram can be plotted; Pálovics, E.; Madarász, J.; Pokol, G.; Fogassy, E.; Bánhegyi, D.F. Economic Separations of Organic Acidic or Basic Enantiomeric Mixtures—A Protocol Suggestion. *Int. J. Mol. Sci.* **2023**, *24*, 846. https://doi.org/10.3390/ijms24010846

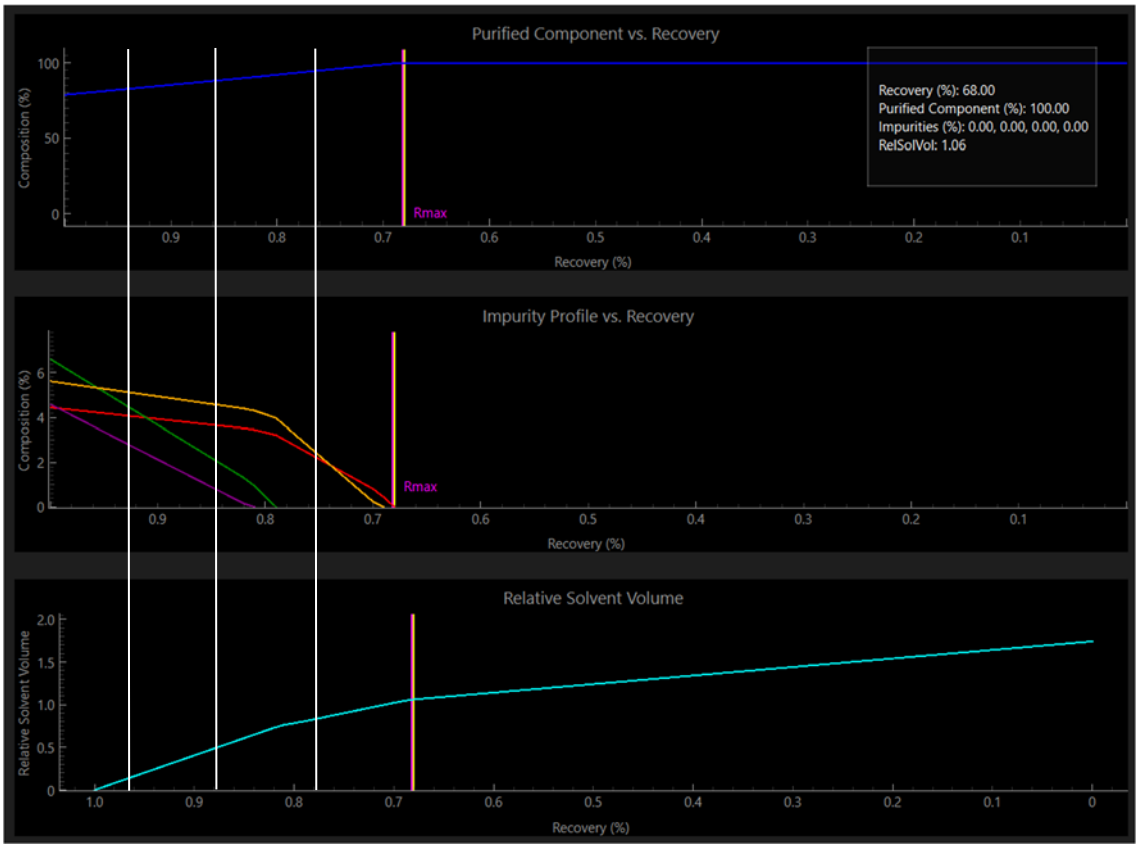


Figure 2 NODES simulation of a 5-component system; the 3 white lines these slices at different Recoveries (R) which correlate with having a different solvent volume (bottom plot) and result in variation of the purity of the target compound (top plot); note the way the impurities vary in a complex way due to their underlying PhysChem parameters. The maximum yield for the system when optimized is 68% physical recovery of pure component 1 (R_{max} = 68%), this the only point in the system where the liquor compositions is the eutectic composition, as distinct from a binary system where the liquors have the eutectic composition throughout the range R = 100% - R_{max}.

3) Understanding Multicomponent Eutectics

In an n-component system:

- The most soluble portion is a eutectic containing all n components.
- Once dissolved, the next most soluble portion is a eutectic of (n–1) components.
- This continues until the least soluble portion remains.

Typically, the least soluble component is the desired product—but occasionally a recalcitrant impurity is least soluble, leading to purification of the wrong species. This is disastrous in practice.

Because the starting composition is known, applying even a single solvent aliquot yields a composition slice at a specific recovery point (white lines in Figure 2). Using a mechanistic equilibrium model, CRYSS fits the data and projects forward to calculate R_{max}.

The power of the method is its scalability: CRYSS can process large datasets automatically, ranking systems from high-yielding to poor. This enables standardized, rapid triage of large screening arrays and selection of the most promising systems for optimization.

In principle, if the fitted parameters are accurate, a single well-constructed experiment is sufficient to estimate R_{max}. Additional solvent volumes and replicates strengthen the dataset and improve confidence.

Performance of CRYSS

Using the NODES App, we can generate arbitrarily large *in silico* datasets that act as proxies for real experimental data obtained from the protocol described earlier. This allows us to choose sensible slice points within the critical composition space, between R = 100% → R_{max}.; this is the eutectic domain. Each slice point corresponds to a specific solvent volume in a real experiment. A single slice typically lies near the middle of this range; multiple slices are distributed evenly, with slight randomization to minimize bias.

NODES can generate mathematically closed, precise equilibrium data using different starting compositions, different PhysChem parameter sets, and optional 1:1 addition compounds. These datasets are internally self-consistent, and—critically—we know the true R_{max} for each mixture. This makes them ideal for demonstrating CRYSS. Beyond perfect numerical data, NODES also includes options to introduce realistic experimental error:

1) Analytical error
    - Capped at 2%
    - Applied with a normal distribution across all components in the initial mixture, solid phase, and liquor phase.
2) Systematic sampling error
    - Simulates cross-contamination of solid into the liquor measurement and liquor into the solid measurement.
    - Models the most common real-world source of error during sampling.

We generated a dataset of 50 mixtures, each with 1–3 slices (proxies for solvent volumes). The files are:

- 260902_1814_Vol01.xlsx
- 260902_1814_Vol01_err5pc.xlsx
- 260902_1814_Vol02.xlsx
- 260902_1814_Vol02_err5pc.xlsx
- 260902_1814_Vol03.xlsx
- 260902_1814_Vol03_err5pc.xlsx

These datasets are available for review here.

CRYSS Performance Results

Single-Volume Data (1 R-slice)

Testing CRYSS with 260902_1814_Vol01.xlsx yields the performance shown in Figure 3, with: R^2 = 0.756. This level of predictivity is already highly acceptable. Even at a single solvent volume, the scatter plot clearly allows rapid visual identification of the best candidates for further experimentation.

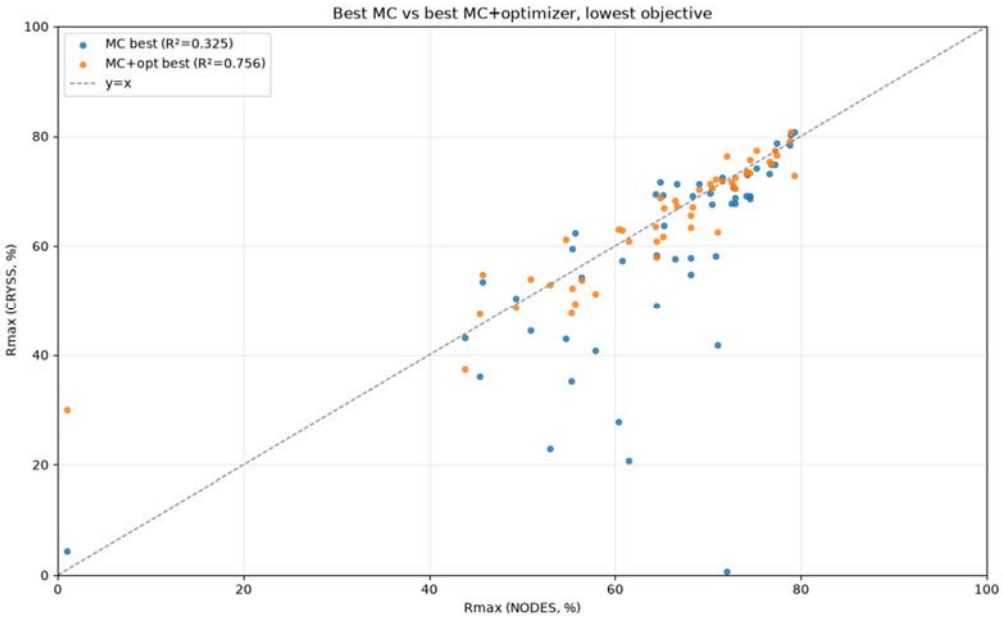


Figure 3: CRYSS run using NODES data from a single solvent volume (one R-slice). The plot compares known NODES R_{max} values with CRYSS-predicted R_{max} values. Blue dots: Monte Carlo simulation only, Orange dots: Hybrid approach (Monte Carlo results fed into an optimizer; best result plotted)

Three-Volume Data (3 R-slices)

Jumping to 260902_1814_Vol03.xlsx, CRYSS achieves: $R^2 = 0.993$. This is shown in Figure 4, demonstrating near-perfect correlation between true and predicted R_{max} values.

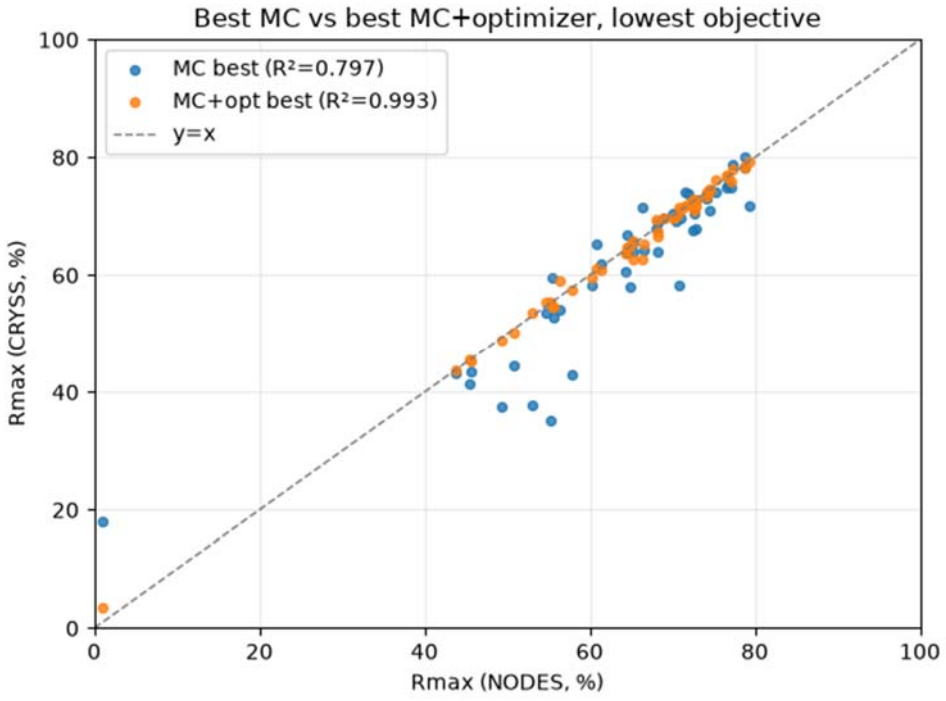


Figure 4: CRYSS run using NODES data from three solvent volumes (three R-slices). Blue dots: Monte Carlo only. Orange dots: Hybrid Monte Carlo + optimizer best result.

Summary Across All Datasets

We repeated this analysis across all NODES test files. The results are summarized in Figure 5. A key observation emerges:

- Experimental error (5%) only becomes visible in the most constrained three-volume datasets.
- For one or two solvent volumes, the inherent noise from fitting dominates, effectively masking the added experimental error.

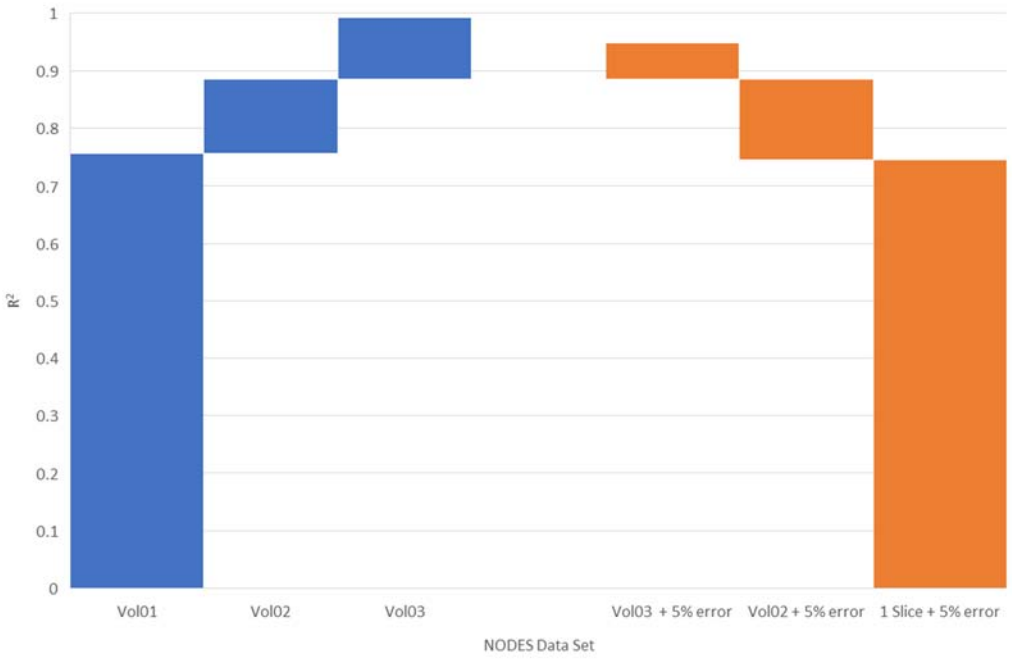


Figure 5: R^2 values for all NODES simulation datasets, comparing optimized results and the incremental improvement gained from additional solvent volumes (more R-slices). Blue: Error-free datasets. Orange: Datasets with 5% error applied

Given that Figure 3 already shows clear visual separation of good *vs*. poor systems, these results indicate meaningful robustness in the CRYSS methodology.

Technical Approach

CRYSS employs a hybrid strategy combining Monte Carlo simulation, which explores the parameter space broadly, and focused optimization, which refines the best Monte Carlo candidates.

CRYSS V1.0 is a GUI-free application. It:

1. accepts Excel input files containing composition data
2. parses replicates& clusters experiments by starting mixture composition
3. identifies R-slices (solvent volumes) via lever-rule analysis with error/outlier correction
4. if a NODES dataset is provided, CRYSS plots R_{max}(NODES) vs R_{max}(CRYSS)
5. solves for R_{max} accordingly and writes results in an Excel file.

CRYSS V2.0 preserves V1.0 underlying mechanistic model but introduces four major advancements:

1. **Enhanced Input Parser** Updated logic supporting the CRYSS V2.0 workbook format, including improved slice handling, replicate absorption, salt-former tracking, and tolerance-based filtering.
2. **Interactive Parser-Validation GUI** A new front-end interface enabling users to tune analytical and noise tolerances, inspect and refine through parser diagnostics, and ensure high-quality data is passed to the optimizer.
3. **CRYSS-P Core (C++ Engine)** A full C++ implementation of the Monte Carlo + optimizer pipeline, fully utilizing multi-core, multi-thread processors to deliver ~40× faster execution, reducing runtime from minutes to seconds.
4. **Interactive Results Viewer** A back-end GUI providing heat map visualization, best-system plot, and interactive stacked composition/solubility analysis (Figure 6 & Figure 7).

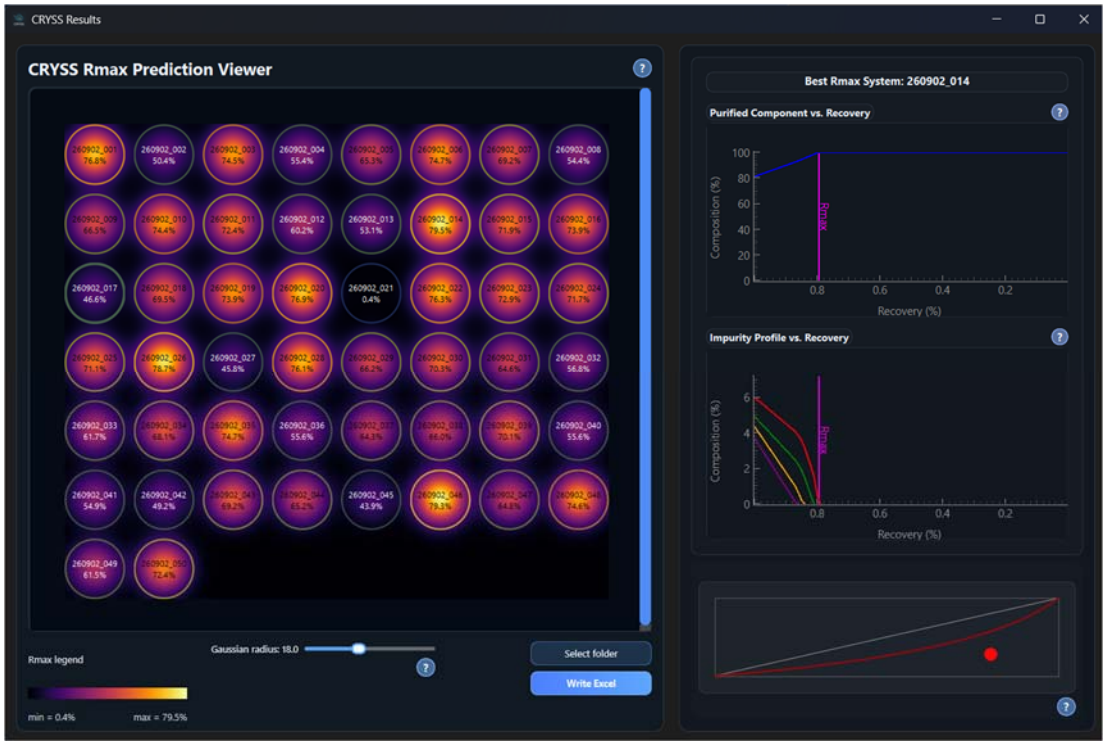


Figure 6: Interactive CRYSS R_{max} Predictions Viewer GUI from CRYSS V2.0 application, showing all results from the simulation, with the best R_{max} system plotted to the upper right



Figure 7: From the CRYSS V2.0 CRYSS R_{max} Predictions Viewer GUI, double-clicking of any heat map well brings up an interactive composition and relative solvent volume stacked plot allowing inspection of the change in state as function of mass partitioned between the solid and solution phase at equilibrium

Expected Impact

CRYSS introduces a fundamentally new capability into crystallization development: the ability to convert small-scale equilibrium partitioning data into mechanistic predictions of maximum achievable yield (R_{max}) across multicomponent mixtures. This transforms early-stage purification screening from a largely empirical exercise into a quantitative, model-driven workflow.

The strength of CRYSS lies in its independence from specific physical property data; instead, these properties are encoded within the observed purification behaviour. The methodology employed in CRYSS makes use of fitting a thermodynamically relevant model to a real system. The primary feature of CRYSS is that captures the actual solubility behaviour of the specific solvent system by absorbing deviations from ideality into the fitted model.[2] This approach provides a realistic estimation of R_{max} in real-world chemical systems.

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[2] The Thermodynamic Solubility Equation: $a_i = \gamma_i . x_i = e^{Q}$; where $Q = \frac{\Delta H_f}{R}\left(\frac{1}{T_f} - \frac{1}{T}\right)$, a_i: is the activity of the solute in the solution, γ_i: is the activity coefficient of the solute in the solvent, x_i: is the mole fraction of the solute in the solution, ΔH_f= is the enthalpy of fusion (melting) of the pure solid solute, T_f= is the melting temperature (in Kelvin) of the pure solid solute, R= is the ideal gas constant, T= is the absolute temperature (in Kelvin) of the system. Thus, in the CRYSS approach we are using real, non-ideal, solubility behaviour of a system we are simply fitting the lumped terms in the exponential; since the measured mass fraction (m_i) is proportional to the mole fraction (i.e. $m_i = \beta_i . x_i$), and therefore, $m_i = e^{Q + \ln(\beta_i) - \ln(\gamma_i)}$, this is how CRYSS is capturing the real system information supplied from analytical measurements.

By ranking solvent, solvent-mixture, and salt-formation systems based on their predicted purification ceiling, CRYSS enables process chemists to focus effort where it matters most. Instead of optimizing dozens of marginal systems, teams can rapidly triage large screening arrays and concentrate resources on the top-performing 20–25% of candidates. This reduces experimental burden, accelerates development timelines, and increases the likelihood of selecting a high-yielding crystallization route early in the project.

The hybrid Monte Carlo + optimization engine provides robustness against noise, allowing meaningful predictions even from minimal data (a single solvent volume per mixture). As a result, CRYSS supports a two-stage experimental strategy: broad, inexpensive initial screening followed by targeted, high-resolution follow-up studies. This approach is well aligned with modern parallel reactor platforms and high-throughput workflows.

Because CRYSS is mechanistic rather than empirical, it scales naturally to complex mixtures, salt-forming systems, and derivatization strategies. CRYSS also integrates seamlessly with NODES, enabling *in silico* exploration of purification landscapes gaining familiarity and assessing experimental accuracy expectations before committing laboratory resources. The source code is already suitable for API integration into automated screening hardware, ELNs, and laboratory informatics systems (this is at release V1.0 level), and CRYSS 2.0 is a GUI enabled efficient user-friendly application which should accelerate process development and optimization is laboratories everywhere.

Taken together, CRYSS has the potential to reshape how crystallization screening is performed: from trial-and-error exploration to predictive, data-driven decision-making.

Practical Considerations

CRYSS is compatible with a range of experimental workflows, from small-scale screening to larger-scale process development. In screening mode, where experiments are typically performed in parallel reactor blocks using sealed vials with septa, the most efficient approach is to sample only the solution phase. Inline filtration or brief centrifugation ensures a clear supernatant. A simple **w/w determination**, combined with known solvent and starting masses, allows calculation of the mass dissolved in the solvent; by difference, the remaining mass in the solid phase is obtained. This directly provides the recovery at the sampling point. Because the starting mixture composition is known, the **lever rule** can be applied to calculate the composition of the solid phase without requiring direct solid analysis, assuming the system is chemically stable under the experimental conditions.

For larger-scale or process-optimization studies—whether using a single reactor or larger parallel units—it may be practical to collect slurry samples, followed by filtration to obtain both liquor and solid compositions. This dataset corresponds directly to the CRYSS input format. In this workflow, CRYSS internally calculates the recovery from the measured compositions, providing a consistent and mechanistically grounded basis for R_{max} prediction and solubility modelling.

Conclusion

CRYSS demonstrates that multicomponent crystallization behaviour — long considered too complex for practical mechanistic analysis — can be decoded, modelled, and used to guide purification strategy. The results from extensive NODES benchmarking show that even sparse equilibrium data contain

enough information to estimate R_{max} with high fidelity, and that hybrid Monte Carlo + optimization methods deliver near-ideal predictivity when multiple solvent volumes are available.

The methodology is robust, scalable, and experimentally efficient. A single well-constructed equilibrium experiment can often provide a good directional estimate of R_{max}, and additional solvent volumes further refine the prediction. CRYSS therefore offers a practical route to mechanistic insight without imposing heavy experimental overhead.

Most importantly, CRYSS provides clarity at the earliest and most consequential stage of crystallization development: selecting which systems deserve attention. By identifying high-yielding solvent and salt-forming conditions before optimization begins, CRYSS helps ensure that development resources are spent on the right problems, not the wrong ones. CRYSS validates a concept that equilibrium partitioning data, interpreted correctly, can reveal the purification limits of complex mixtures.

CRYSS stands as a practical, mechanistic, and experimentally grounded tool that brings predictive power to crystallization screening — and opens the door to a more rational, data-driven future for purification science.

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